Glycobiology, 2024, **34**, cwae005
https://doi.org/10.1093/glycob/cwae005
Advance access publication date 22 January 2024
Original Article

Studying the *O*-GlcNAcome of human placentas using banked tissue samples

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***O*-GlcNAcylation is a dynamic modulator of signaling pathways, equal in magnitude to the widely studied phosphorylation. With the rapid development of tools for its detection at the single protein level, the *O*-GlcNAc modification rapidly emerged as a novel diagnostic and therapeutic target in human diseases. Yet, mapping the human *O*-GlcNAcome in various tissues is essential for generating relevant biomarkers. In this study, we used human banked tissue as a sample source to identify *O*-GlcNAcylated protein targets relevant to human diseases. Using human term placentas, we propose (1) a method to clean frozen banked tissue of blood proteins; (2) an optimized protocol for the enrichment of *O*-GlcNAcylated proteins using immunoaffinity purification; and (3) a bioinformatic workflow to identify the most promising *O*-GlcNAc targets. As a proof-of-concept, we used 45 mg of banked placental samples from two pregnancies to generate intracellular protein extracts depleted of blood protein. Then, antibody-based *O*-GlcNAc enrichment on denatured samples yielded over 2000 unique HexNAc PSMs and 900 unique sites using 300 μg of protein lysate. Due to efficient sample cleanup, we also captured 82 HexNAc proteins with high placental expression. Finally, we provide a bioinformatic tool (CytOVS) to sort the HexNAc proteins based on their cellular localization and extract the most promising *O*-GlcNAc targets to explore further. To conclude, we provide a simple 3-step workflow to generate a manageable list of *O*-GlcNAc proteins from human tissue and improve our understanding of O-GlcNAcylation's role in health and diseases.**

Key words: database; O-GlcNAc; placenta; proteomic; STRING.

Over the past 40 years, O-GlcNAcylation has emerged as a significant post-translation modification of proteins, regulating over 8,000 human proteins (Wulff-Fuentes et al. 2021). O-GlcNAcylation is an atypical glycosylation that involves the addition of a single GlcNAc molecule to serine or threonine sites on proteins through an *O*-linked β-glycosidic bond. The addition is catalyzed by O-GlcNAc transferase (OGT), and O-GlcNAcase (OGA) catalyzes the hydrolytic cleavage. Additionally, O-GlcNAcylation is a product of nutrient flux through the hexosamine biosynthetic pathway (HBP), using the donor substrate uridine diphosphate GlcNAc (UDP-GlcNAc). Over the years, advancements in the O-GlcNAc toolbox have led to a better understanding of its signaling roles in various tissues. First, the development of multiple antibodies recognizing the O-GlcNAc modification on proteins, such as RL2 and CTD110, have allowed the observation of global O-GlcNAcylation changes in various physiological states such as in response to stress (Fahie et al. 2022) or during the cell cycle (Saunders et al. 2023). However, the need for more targeted approaches to study site glycosylation changes on single proteins rapidly arose as O-GlcNAc's roles in diseases were discovered. The deregulation of O-GlcNAcylation has been closely linked to several human diseases, including, but not limited to, cancer, diabetes, and Alzheimer's. In diabetes, one way this PTM takes effect is hyperglycemia, which leads to an increase in OGT production, which then causes excessive O-GlcNAcylation (Ma and Hart 2013). This prolonged hyper-GlcNAcylation results in impaired insulin secretion and pancreas apoptosis (Ma and Hart 2013). Our lab previously showed a differential expression of *OGT* in the placenta from Gestational diabetes patients and demonstrated that O-GlcNAcylation can impact placental hormone secretion (Cui et al. 2023).

Furthermore, others have shown that reduced *OGT* is a marker of gestational stress, and this reduction is exacerbated in the placentas of male fetuses (Howerton et al. 2013; Wang et al. 2022). However, the O-GlcNAcome of the human placenta has yet to be explored. This article provides proof of concept to study the O-GlcNAcome of the human placenta.

Indeed, knowing the sites and specific protein targets for O-GlcNAcylation is highly promising to serve as biomarkers or therapeutic targets, and subsequent tools were built with that idea in mind, such as nanobodies and aptamers (Ramirez et al. 2021; Zhu and Hart 2023). Both approaches aim to target specific protein O-GlcNAcylation and de-O-GlcNAcylation with the distant goal of therapeutic intervention. Currently, O-GlcNAc drugs like Thiamet-G (OGA inhibitor) are being tested in clinical trials to impact global O-GlcNAcylation in the brain (Bartolomé-Nebreda et al. 2021). However, future targeted approaches are more specific and require information on the sites/proteins modified by O-GlcNAcylation and the role this modification plays on the protein of interest.

Received: December 23, 2023. **Revised:** January 11, 2024. **Accepted:** January 11, 2024
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Advances in mass spectrometry (MS) over the years and the generalization of EThcD MS for PTMs has tremendously increased the detection and site-mapping of *O*-GlcNAcylated proteins (reviewed in (Burt et al. 2022)). Due to the increasing amount of *O*-GlcNAcomic data from these proteomic studies, we created the *O*-GlcNAc database to provide an online platform (oglcnac.mcw.edu) to browse published *O*-GlcNAc proteins. This database is a good starting point for scientists wishing to study a specific protein (Wulff-Fuentes et al. 2021).

As highlighted by the database and the meta-analysis of the *O*-GlcNAcome (Wulff-Fuentes et al. 2021), some proteins are more abundantly *O*-GlcNAcylated and are usually found in almost every *O*-GlcNAcomic study. These proteins, such as Sp1, HCF1, nucleoporins, etc., are ubiquitous and make poor therapeutic targets as they are essential in all cells and tissues. On the contrary, tissue/cell-specific proteins with fewer *O*-GlcNAcylation sites and more specific functions make better candidates. However, finding these proteins and mapping their sites is challenging (Burt et al. 2022). Indeed, *O*-GlcNAcylation is a sub-stoichiometric (under 20% occupancy) modification for most proteins (Rexach et al. 2012; Darabedian et al. 2018; Leturcq et al. 2018). Thus, it is necessary to enrich *O*-GlcNAcylated proteins before analysis by MS to discover better biomarker candidates in diseases.

To attain this goal, multiple approaches were developed, including lectin pull-down with Wheat Germ Agglutinin (WGA) or chemical labeling of *O*-GlcNAcylation with clickable GalNAz, thus allowing the attachment of various affinity handles for pull-down (Trinidad et al. 2012; Ma and Hart 2014; Burt et al. 2022). Due to the poor performance of the current *O*-GlcNAc antibody RL2 and CTD110, immunoprecipitation has never been widely used to enrich *O*-GlcNAcylated proteins until recently. Indeed, a new set of 4 antibodies (PTMScan *O*-GlcNAc [GlcNAc-S/T] Motif Kit #95220, Cell Signaling Technology, Inc) were designed to enrich for *O*-GlcNAcylated proteins (Burt et al. 2021). Applying this antibody-based enrichment strategy to mouse brain tissue samples identified over 1,000 unique *O*-GlcNAc-modified peptides and sites using a smaller sample size and instrument time than previously published (Burt et al. 2021). Moreover, compared to WGA pulldown, this approach was more specific to single HexNAc modification rather than extended glycans (2xHexNAc), which are characteristic of other types of extracellular *O*-linked glycans. Compared to chemo-enzymatic labeling, the significant advantage of this approach is the minimization of the number of experimental steps before MS, thus increasing the yield of GlcNAc identification.

In this study, we expand on this significant advancement in *O*-GlcNAc enrichment protocols by adopting this approach to human-banked samples, which often presents substantial challenges. Patient samples are often banked without proper washing, thus contaminating cells with abundant *O*-glycosylated blood protein. These glycoproteins, such as human serum albumin, antibodies, and hemoglobin, pollute MS acquisition, masking the less abundant intracellular proteins and, even more, their *O*-GlcNAcylated forms.

Thus, using freshly banked placentas, we designed an optimized method to prepare human placental samples with naturally high blood content (Kruger et al. 2023). We then enriched for *O*-GlcNAcylated protein using the PTMscan antibodies previously mentioned (Burt et al. 2021). Our optimized protocol yielded over 60% of HexNAc on total peptides. Per sample, we obtained up to 2,600 HexNAc-containing PSMs, 1,090 unique HexNAc peptides, and more than 1,000 unique HexNAc proteins using 45 mg of starting placenta. Furthermore, we present a bioinformatic workflow to sort these HexNAc PSMs and extract the most confident *O*-GlcNAcylated proteins and sites. Thus, we confidently identified 641 *O*-GlcNAcylated proteins, including up to 146 new proteins and 730 new sites within the two placental samples analyzed. Finally, we enriched over 50 high-placental-expression O-GlcNAcylated proteins representing new biomarkers of placental *O*-GlcNAcylation.

Overall, this study provides a framework for mapping and analysis of the O-GlcNAcome from banked patient samples.

Results

Preparation of freshly banked placenta tissues

Banked placental tissues present unique challenges and require additional pre-processing steps. Based on the institution, tissue banking might not include sample washing before flash freezing. For the placenta, which is naturally high in blood content, this leads to extensive amounts of serum albumin, antibodies, and other O-linked glycosylated proteins, including *O*-linked GalNAc, indistinguishable in MS from *O*-linked GlcNAc. Furthermore, *O*-GalNAc, found in the extracellular or membranous compartment, are more abundant than *O*-GlcNAc proteins found in the cytosol, nucleus, or mitochondria and decrease our discovery rate for *O*-GlcNAc proteins. Thus, the goal of the pre-processing steps is to clear the sample of (1) extracellular and membranous proteins that contain *O*-GalNAc modification and (2) abundant blood proteins.

To clear the sample of extracellular and membranous proteins, the banked tissues were lysed and centrifuged to pellet the membrane fraction (Fig. 1A). To remove highly abundant blood proteins from the supernatant, we tested two different types of High-Select depletion columns (Thermofisher Scientific): "Human Serum Albumin (HAS)/Immunolglobulins (Igs)" and "Top 14 abundant proteins". MS analysis of two placenta samples after "Human Serum Albumin (HSA)/ Immunoglobulins (Igs)" or "Top14 abundant proteins" High-Select depletion columns show a complete absence of PSM associated with HSA and Igs (Fig. 1B). However, with the High-Select Top 14 abundant proteins, other highly abundant blood proteins were significantly depleted (Fig. 1B). Additional blood proteins (blood markers extracted from (Kruger et al. 2023)) (Table S1) were still found in the sample run in the Top 14 Abundant proteins column; however, low PSM counts suggested that it would not impact the detection of lower abundance *O*-GlcNAcylated proteins (Fig. 1B, Table S1).

To visualize the result of each depletion, samples were concentrated by chloroform/methanol protein precipitation and loaded on SDS-PAGE. Total Protein Staining (TPS) after western blotting on the depleted samples showed a significant reduction of abundant proteins in the lysate without depleting cytosolic proteins, such as actin (Fig. 1C). As a result of depletion, using the "Top 14 abundant proteins" column significantly increased the ratio of placental markers to blood markers compared to the HSA/Igs depletion column (Fig. 1D and E, Table S1).

Finally, following these pre-processing steps, we assessed the subtype of placental cells in our samples (Fig. 1F). Using placental markers described previously (Kruger et al. 2023),

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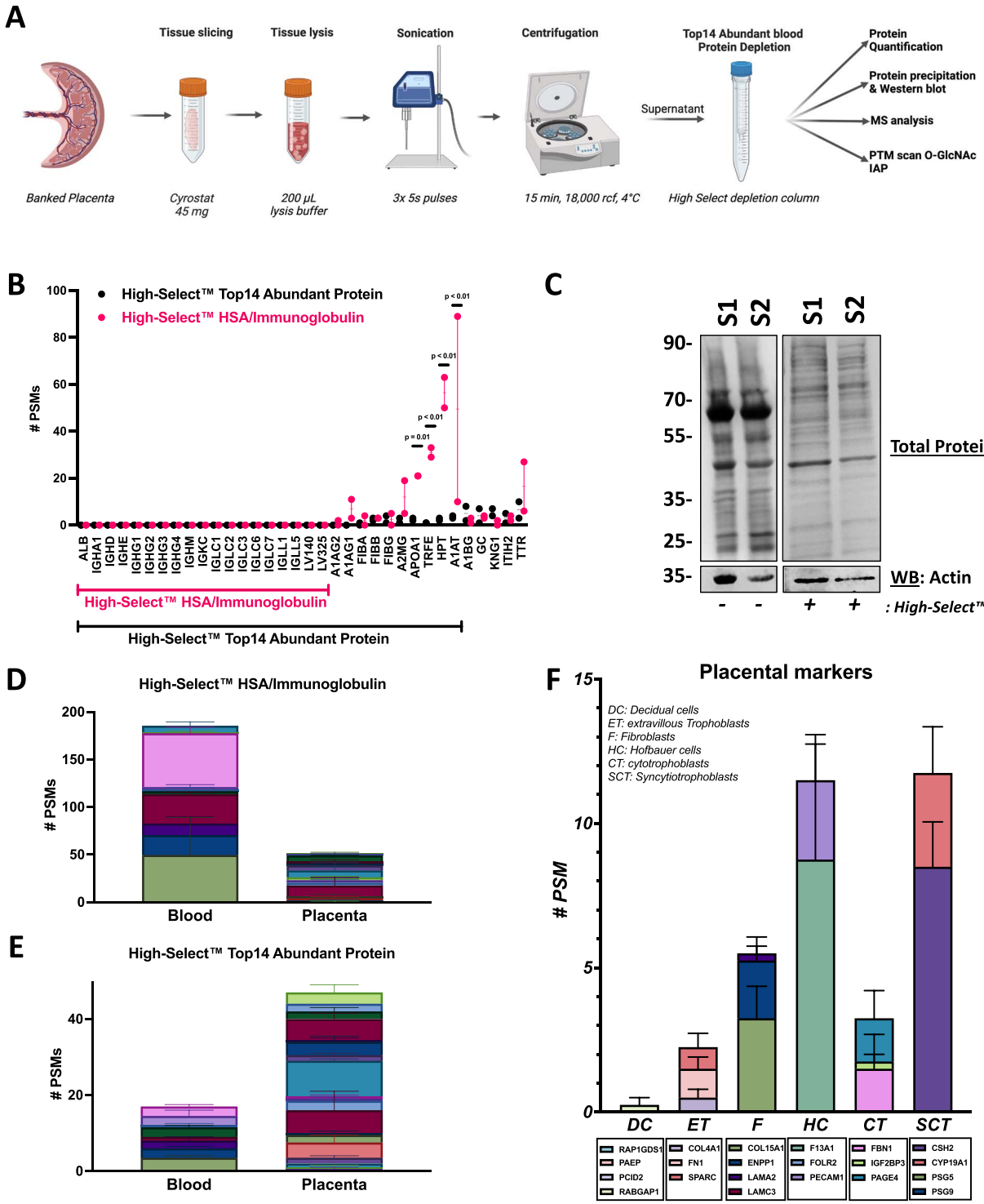


Fig. 1. Preparation of freshly banked placenta tissues. A) Protein preparation workflow. B) PSM counts for blood proteins using two different high-select depletion columns. C). Western blot of Total Protein Stain and Actin for samples 1 and 2 pre-(−) and post-(+) depletion column. D/E) PSM counts for blood and placental proteins after the High-select Human Serum Albumin (HSA)/Immunoglobulin (D) or Top 14 abundant protein depletion column (E). F) PSM counts for placental markers organized by placental cell type. Significance was measured by Student T-Test. *n*= 4 technical replicates, *n* = 2 samples.

we observed that the samples contained larger PSM counts of markers from syncytiotrophoblasts (SCT), the nutrient-sensing endocrine cells of the placenta, and Hoffbauer Cells (HC), villous macrophages of fetal origin. We also observed significant PSM counts for Cytotrophoblasts (CT), precursor to syncytiotrophoblasts, extravillous trophoblasts (ET) that invade and remodel the uterine wall and vessels, and Fibroblasts (F). All the latter emerge from the fetal side, suggesting that we recovered a significant amount of the fetal placenta. However, limited markers for decidual cells of maternal origins were recovered. Therefore, our protocol successfully removed contaminating proteins of blood origin,

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with limited impact on the abundance of proteins from the target placental tissue.

In summary, starting with 45 mg of tissue, ~300 μg of proteins was recovered after the Top14 abundant proteins High-Select depletion column and was elected for further enrichment protocols.

Optimization of *O*-GlcNAc immunoaffinity purification (IAP) protocol on full-length proteins

The next step of identifying relevant *O*-GlcNAcylated targets in the placenta was enriching for *O*-GlcNAcylated proteins. We took advantage of a recently published immunoaffinity purification (IAP) protocol utilizing a set of 4 *O*-GlcNAc antibodies bound to beads (commercialized by Cell Signaling under the name PTM-Scan *O*-GlcNAc), which had shown promising results (Burt et al. 2021). Following this enrichment protocol, EthcD MS/MS was performed to measure HexNAc-modified peptides (Fig. 2A, Right panel, Tables S2–S5). IAP led to a slight increase in the HexNAc enrichment ratio (calculated by dividing the HexNAc PSMs by total PSMs), increasing from 3% in the Whole Tissue Lysates (WTL) to 8% in the IAP (Fig. 2B, Table 1, Tables S2–S5). The enrichment ratio was greater when looking at unique peptides or proteins (Fig. 2C and D, Table 1, Tables S2–S5). However, HexNAc PSMs (S1: 139; S2: 169) following IAP were significantly lower than WTL (S1: 1,007; S2: 832), which was concerning because it indicated that many unmodified peptides were co-purified, which increased unwanted peptide backgrounds in MS (Fig. 2E). We also noticed that even in the original publication (Burt et al. 2021), HexNAc peptides were only enriched at a ratio of up to 25%. Thus, to improve the enrichment ratio, we modified this protocol as presented in Fig. 2A (Right panel) (Tables S6–S9). In this optimized protocol, we reduced and alkylated the proteins from WTL, leading to unfolded, accessible *O*-GlcNAc epitopes for the antibodies to bind. Then, we purposefully did not digest the sample with protease (e.g. trypsin) but rather pursued with full-length, denatured proteins. The undigested protein samples were cleaned in between steps with a chloroform/methanol purification rather than C18 column for peptide cleanup in the original protocol using digested lysates. Samples were finally digested with trypsin before MS analysis. This optimized protocol yielded more than 2,100 HexNAc PSMs and an enrichment ratio of over 65% (Fig. 2B and E, Table 1, Tables S6–S9). The high enrichment ratio was maintained when looking at peptides and proteins with over 830 unique HexNAc peptides and 780 unique HexNAc proteins identified (Fig. 2C and D, Table 1, Tables S6–S9), with the two samples overlapping at around 19% at the HexNAc-containing peptide level and 23% overall. A similar overlap (17%) with the original protocol confirmed great diversity between the two human placenta samples analyzed (Tables S2–S5). Thus, we confidently identified 908 unique HexNAc sites using 300 μg of starting protein sample (Table S10).

Working with undigested lysate also allows the visualization of enrichment on SDS-PAGE. Thus, WTL and IAP samples were labeled with the click-It *O*-GlcNAc labeling kit combined with TAMRA detection to visualize *O*-GlcNAcylated proteins using purified crystallin as a positive control. (Fig. 2F). This protocol was specific for O-GlcNAcylated proteins as indicated by no staining in Click-It labeling reactions performed without the GalNAc Transferase (Fig. 2G). Both samples were significantly enriched for O-GlcNAcylated proteins as indicated by an increase in total abundance and enrichment of unique bands in the PTM Scan lanes compared to WTL lanes.

To conclude, we significantly enhanced the enrichment protocol of HexNAc peptide using antibody-based purification, leading to the identification of over 900 unique hexNAc sites with only 300 μg of starting protein lysates from cleaned human banked tissues.

Bioinformatic sorting of HexNAc proteins.

One of the caveats of MS analysis for mapping *O*-GlcNAcylated proteins is that *O*-GlcNAc carries the same mass as *O*-GalNAc, a signature of extracellular *O*-linked glycans. Indeed, we observed extracellular proteins, including mucins and secreted growth factors, in our datasets. Thus, post-processing of the MS data is necessary to sort out the two different kinds of glycosylation. It is possible to distinguish between the two sugars by analyzing the Oxonium ion (Li et al. 2022). However, this process requires extensive data analysis, which can be time-consuming and labor-intensive. Thus, we aimed to facilitate sorting MS data generated with *O*-GlcNAcomic approaches using cellular localization to differentiate between *O*-GalNAc and *O*-GlcNAc found in extracellular and intracellular compartments, respectively (Fig. 3A).

Thus, we utilized the STRING database through Cytoscape to extract the cellular localization of each identified protein. First, UniProtKB Identifiers (ID) were entered into Cytoscape using the STRING protein search mode, which gathered cellular compartment information. This initial analysis showed that the optimized protocol significantly enriched cytosolic and nuclear protein (Fig. 3B). However, many proteins also showed extracellular compartmentalization (either in the plasma membrane or extracellular compartment) (Fig. 3B). While significantly reduced in IAP compared to WTL, this observation prompted an extensive sorting of the HexNAc-containing proteins identified by MS to extract the best *O*-GlcNAc candidates.

Therefore, we created a small tool, CytOVS, to automatically import the HexNAc proteins and their corresponding MS data (e.g. number of PSM and HexNAc) into Cytoscape and group them by the likelihood of being an authentic O-GlcNAc protein (e.g. *O*-GlcNAc probability). This is done by extracting the compartment information for each protein through the STRING network linked to the COMPARTMENT database (Binder et al. 2014; Doncheva et al. 2023). This database maps all evidence to protein identifiers such as UniProtKB ID and assigns confidence scores for each cellular localization from 0 (none) to 5 (high confidence). The network generated with CytOVS for sample 1 IAP is provided in supplementary documents.

Thus, cellular compartment information was gathered to identify the proteins carrying HexNAc that are extremely extracellular (>4.5), which were sorted into the "Extracellular HexNAc" group (Fig. 3C and D). Some known *O*-GalNAc proteins found in this group were mucins (MUC5B), Growth Factors (EGF), or extracellular matrix proteins (LAMA5) (based on the GlyGen database) (Tables S11 and S12). Proteins with a score of >4 in either nucleus, cytosol, or mitochondria, the three compartments where *O*-GlcNAc have been identified, were classified as the "*O*-GlcNAc" group

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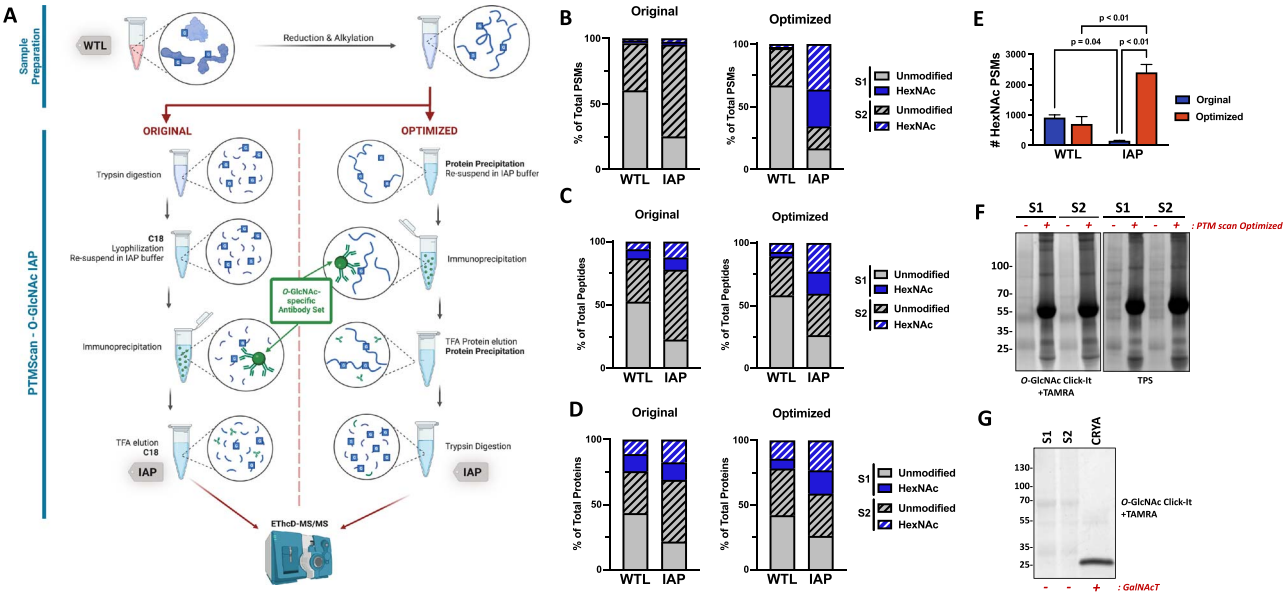


Fig. 2. Optimization of *O*-GlcNAc immunoaffinity purification (IAP) protocol on full-length proteins. A) PTM Scan workflow. B–D) Percentage of HexNAcs and unmodified PSMs (B), peptides (C), or proteins (D) found in Whole Tissue Lysate (WTL) and Immunoaffinity purification (IAP) from the original and optimized protocols. E) Total HexNAcs PSMs found in WTL and IAP samples from the original and optimized protocols. F) *O*-GlcNAc click-it TAMRA Gel and Total Protein Stain (TPS) western blot before and after optimized PTM scan for both samples. G) Negative Control for click-it was performed without GalNAc transferase (GalNAcT). Positive control was performed with purified alpha-crystallin. Both were performed on the same gel but cropped for presentation. Significance was measured by a two-way ANOVA; $n = 4$ technical replicates, $n = 2$ samples.

Table 1. Summary of the PSM, peptide, and protein counts identified in this study using the original and optimized protocols.

Protocol	Sample	# Unmodified PSMs		# HexNAc PSMs		HexNAc/Total PSMs (%)		# unmodified peptides		# HexNAc peptides		HexNAc/Total Peptides (%)		# unmodified proteins		# HexNAc proteins		HexNAc/Total proteins (%)	
		WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP	WCL	IAP
Original	1	31054	1769	1007	139	3%	8%	4415	167	597	71	14%	43%	1860	113	563	71	30%	63%
	2	18804	4713	832	169	4%	4%	2878	408	509	93	18%	23%	1382	249	480	92	35%	37%
Optimized	1	28783	3367	463	2148	2%	64%	4236	1252	264	833	6%	67%	1427	1134	256	787	18%	69%
	2	13573	3948	953	2664	7%	67%	2261	1575	527	1096	23%	70%	1227	1418	485	1009	40%	71%

High to low counts of PSMs, peptides or proteins are depicted on a spectrum of green to red.

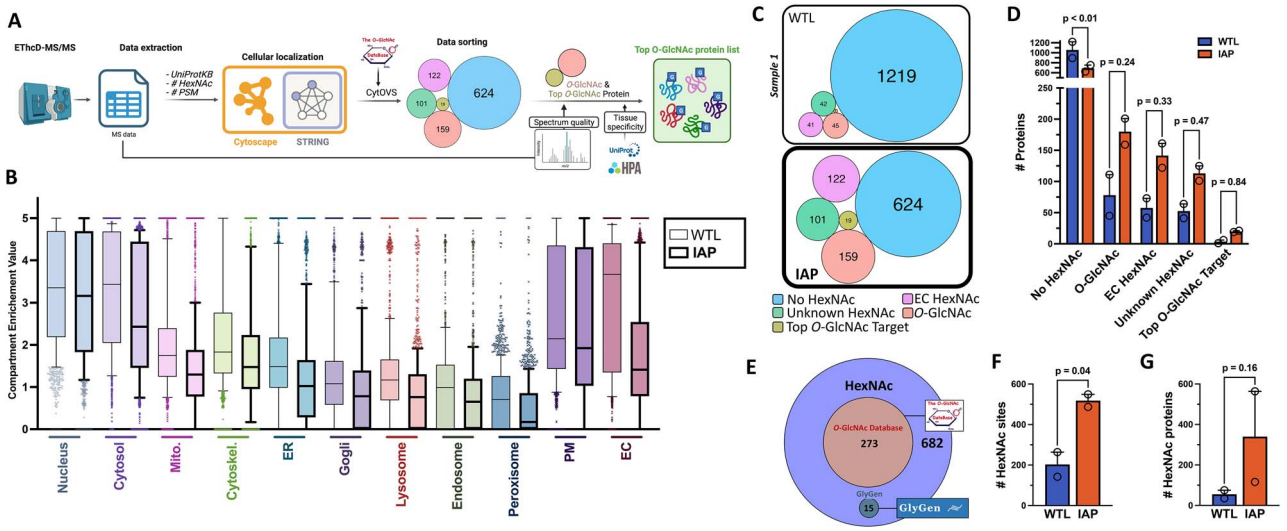


Fig. 3. Bioinformatic sorting of HexNAc proteins. A)**.** Bioinformatics workflow. B) Compartment enrichment for Whole Tissue Lysate (WTL) and Immunoaffinity purification (IAP). Mito: Mitochondria; ER: Endoplasmic Reticulum; PM: Plasma Membrane; EC: Extracellular. C) Number of proteins in each category for sample 1 in WTL and IAP generated with the CytOVS tool. D) Summary of the number of proteins in each category of the CytOVS tool for both samples in WTL or IAP conditions. Significance was measured by a two-way ANOVA. E) Venn diagram of the HexNAc proteins overlapping with the *O*-GlcNAc database and the cataloged *O*-linked glycans in Glygen. F/G) Number of unique HexNAc sites (F) and proteins (G) identified with confidence in both samples in WTL or IAP conditions. Significance was measured by a Student T-Test. $n = 4$ technical replicates, $n = 2$ samples.

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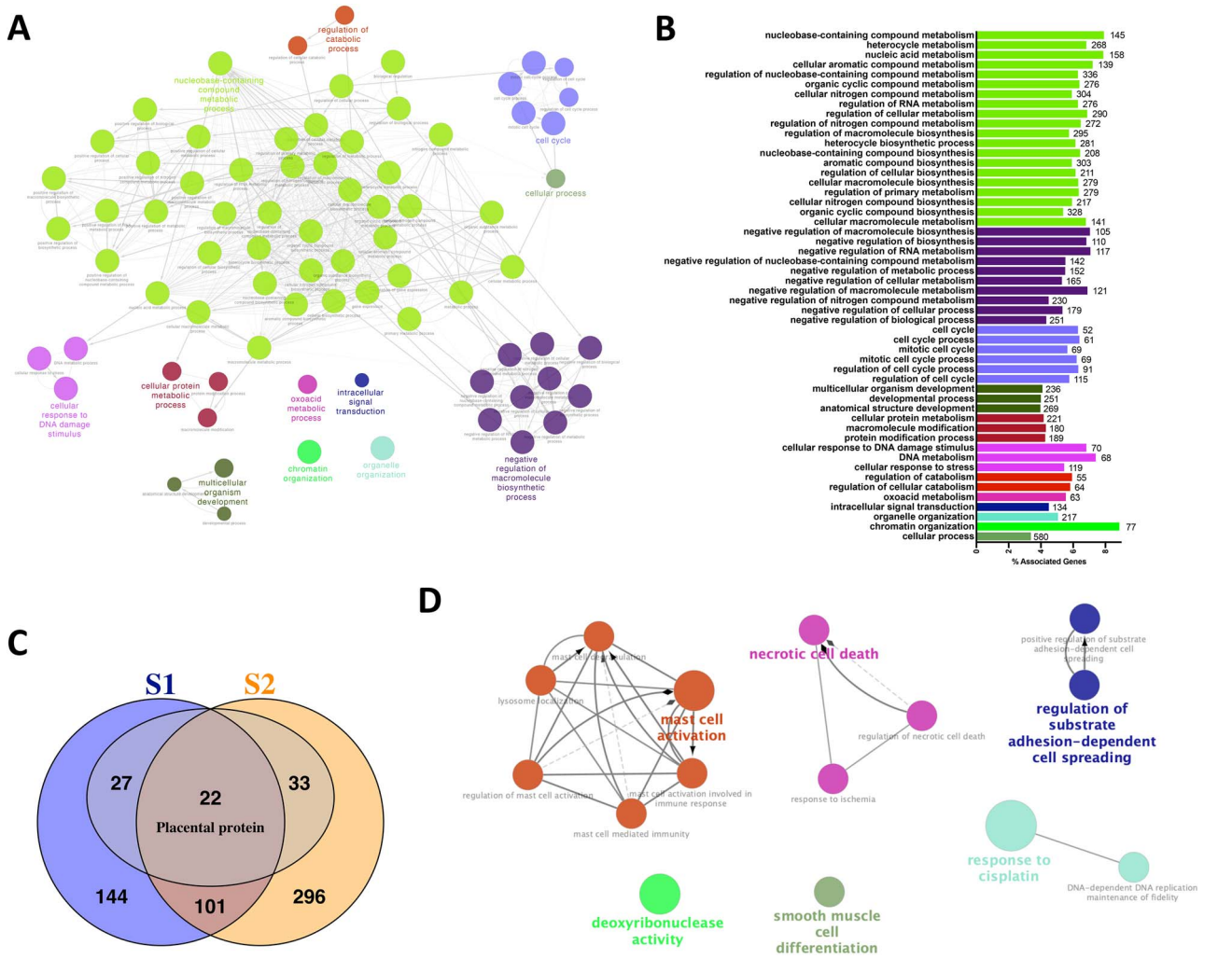


Fig. 4. Functional analysis of *O*-GlcNAcylated proteins. A/B) Gene ontology (GO) analysis of the *O*-GlcNAcylated protein extracted from CytoOVS represented as a network generated on Cytoscape (A) or a diagram representing the percentage of genes found (B). Each bar graph is labeled with each category's total number of genes. C). Venn Diagram representing the overlap for both sample *O*-GlcNAcylated protein (from CytOVS) and their expression in the placenta extracted from UniProtKB. D). GO analysis of the placental *O*-GlcNAcylated protein identified in this study.

(Fig. 3C and D). Some well-characterized/highly studied *O*-GlcNAc proteins found in this group were nucleoporins (Nup214, Nup42), epigenetic regulators (TET1, HDAC6), and transcription factors (Sp1, TFE2) (Tables S11 and S12). An additional category is available for the *O*-GlcNAcylated proteins showing a higher PSM number and will be sorted in the top *O*-GlcNAc protein group. Proteins that carried a HexNAc and did not fit either set were classified as "Unknown HexNAc." While these proteins might be *O*-GlcNAcylated, their localization was too ambiguous to be included in the top *O*-GlcNAc group. Another group, "No HexNAc," was created with the protein that did not contain HexNAc modification (Fig. 3C and D). Using this grouping, we confirmed the significant enrichment of HexNAc-containing groups rather than "No HexNAc" proteins following IAP (Fig. 3C). When considering all HexNAc-containing proteins, we did observe a substantial overlap with protein in the O-GlcNAc database (Wulff-Fuentes et al. 2021) (Fig. 3E) rather than the GlyGen *O*-GalNAc listed protein (York et al. 2020), suggesting that we are aligning with previous *O*-GlcNAcomic studies. Furthermore, using this workflow, we attributed 407/908 unique HexNAc sites to be *O*-GlcNAcylated with high confidence based on their cellular localization, 162/908 to be extracellular and likely to be *O*-GalNAc proteins (Table S10). The remaining 339 were classified as unknown HexNAc because of indecisive cellular localization or unmatched in the STRING analysis (Table S10).

Additionally, CytOVS automatically interfaces with the *O*-GlcNAc database, providing a quick view of the overlap with existing O-GlcNAcomic studies. Thus, we also demonstrated that this protocol led to the confident identification of new *O*-GlcNAcylated proteins and sites. Indeed, from the CytOVS *O*-GlcNAc group, 157 new unique proteins and 234 new unique *O*-GlcNAc sites were confidently identified (Fig. 3F and G) (Wulff-Fuentes et al. 2021).

Functional analysis of *O*-GlcNAcylated proteins

We then selected the *O*-GlcNAc protein from CytOVS and performed a Gene Ontology (GO) enrichment analysis using STRING. We noticed that *O*-GlcNAcylated proteins extracted from this study were involved in regulations of DNA transcription, protein modification, and cell cycle (Fig. 4C and D, Table S13). Previous GO analysis of the

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O-GlcNAcome showed similar enrichment GO terms (Wulff-Fuentes et al. 2021) and confirmed once more the proper grouping of the proteins. In the most enriched GO term in this analysis, new O-GlcNAcylation sites were found at S256 on the RNA trafficking regulator Heterogeneous nuclear ribonucleoprotein A3 (HNRNPA3) and at S139 on the DNA binding protein, Protein cramped-like (CRAMP1) (Fig. S1A and B, Table S10). As part of the insulin signaling pathway, we found a new O-GlcNAcylation site for the X-linked Insulin receptor substrate 4 (IRS4) on S570 and that Phosphatidylinositol 3-kinase C2 domain-containing subunit γ (PIK3C2G) was O-GlcNAcylated at T1232 (Fig. S1C and D, Table S10).

We finally wanted to analyze the tissue specificity, particularly for the placenta, of the O-GlcNAcylated proteins. Thus, we extracted the tissue distribution from UniProtKB for each protein. Respectively, 49/278(S1) and 55/397(S2) proteins were identified as expressed explicitly in the placenta (Fig. 4C). Notably, we identified the Replication factor RPA4 as a new O-GlcNAcylated protein, which was modified in S11 and S13 (Fig. S1E, Table S10). The Nuclear factor of activated T-cells NFATC3 was also captured in this study and, while already present in the database, showed S82 as a new O-GlcNAc site (Fig. S1F, Table S10). All the proteins mentioned in this section were found explicitly in the optimized IAP protocol samples, but none of the other conditions (Fig. S1G). Additionally, GO enrichment of the placental O-GlcNAc proteins showed enrichment of different GO terms than the previous O-GlcNAcome analysis, with specific enrichment in mast cell activation, adhesion, and necrotic cell death (Fig. 4D).

Discussion

O-GlcNAcylation is a dynamic modulator of signaling pathways, equal in magnitude to the widely studied phosphorylation (Bond and Hanover 2013; Hart 2014). With the emergence of PET ligands to detect discrepancies in O-GlcNAc enzymes(Lu et al. 2020), the development of novel inhibitors suitable for human trials(Bartolomé-Nebreda et al. 2021), and the use of nanobodies for drug delivery(Yang et al. 2022), the O-GlcNAc pathway is rapidly emerging as a novel diagnostic and therapeutic target to identify and treat human diseases (Zhu and Hart 2020). However, achieving this goal relies on correctly identifying O-GlcNAcylated proteins and sites that could serve as diagnostic and therapeutic targets. So far, in humans, more than 12,000 serines and threonines on 8,546 intracellular proteins have been post-translationally modified by O-GlcNAcylation (Bond and Hanover 2013; Hart 2014; Wulff-Fuentes et al. 2021). Most of these proteins and sites were mapped in human cell lines and still require *in vivo* validation, as cell lines often represent oversimplified systems that lack biological context. Nevertheless, O-GlcNAcomic studies using human tissues are still rare and are limited to post-mortem human brains (Wang et al. 2017).

One of the reasons for the lack of O-GlcNAc information using human tissue is the difficulty in obtaining sufficient and high-quality tissue. In contrast to biopsy tissue samples, usually only available in small amounts, the placental organ is available in excessive mg amounts at many research institutes. However, although highly abundant, the placenta presents specific challenges for O-GlcNAcomic analysis: its high blood content. This leads to contamination of MS data by (1) peptides of abundant blood proteins such as Human Serum Albumin (HSA) and Immunoglobulins (Igs) and (2) circulating extracellular glycoproteins. This is particularly relevant in O-GlcNAcomic analysis as the O-linked GalNAc core found on these glycoproteins is indistinguishable by MS from GlcNAc (marked as HexNAc), and, thus, lead to false positive identification of O-GlcNAc sites.

To remedy this issue, we have created a workflow that consists of cleaning highly abundant blood proteins through a depletion resin targeting HSA, albumin, IgG, IgA, IgM, IgD, IgE, kappa and lambda light chains, alpha-1-acid glycoprotein, alpha-1-antitrypsin, alpha-2-macroglobulin, apolipoprotein A1, fibrinogen, haptoglobin, and transferrin. Using this depletion method, we detected many PSMs from placental-specific proteins rather than blood, using unwashed placental starting material banked at term.

While cleaning blood from human placental samples was essential, enrichment of O-GlcNAcylated proteins is necessary to discover new O-GlcNAcylated proteins and sites due to the substoichiometric nature of the O-GlcNAc modification. Indeed, while some ubiquitous proteins show almost 100% O-GlcNAcylation at any given moment (e.g. Sp1, Nucleoporins...) and are systematically found in O-GlcNAcomic studies, less abundant and/or less modified and/or tissue-specific proteins are often missed due to the very low abundance of O-GlcNAcylated peptides (Rexach et al. 2012; Darabedian et al. 2018; Leturcq et al. 2018). These proteins, however, are of high value and often make perfect tissue-specific or pathway-specific targets for diagnostics and therapeutic applications.

Although chemoenzymatic labeling combined with Click-chemistry has proven beneficial to enrich O-GlcNAylated proteins and peptides (Trinidad et al. 2012; Ma and Hart 2014; Burt et al. 2022), the number of required experimental steps in this workflow results in protein loss and false positives. Thus, the commercialization of new antibody sets for O-GlcNAcylation immunoaffinity purification (IAP), allowing the direct enrichment of O-GlcNAc peptides, was a significant technical advancement in the field (Burt et al. 2021). This protocol, designed to enrich O-GlcNAc peptides, required trypsinization before IAP. However, in the original publication and our effort with this protocol, many unmodified peptides were found following IAP, suggesting a high level of non-specific binding using this original method. Thus, we demonstrate that denatured proteins rather than digested peptides can reduce non-specific binding in IAP and improve the HexNAc enrichment ratio, reaching more than 60% (HexNAc/unmodified PSMs). Thus, using one placental sample, we identified over 2,100 unique HexNAc PSMs and, up to 1,096 unique HexNAc peptides and 1,009 proteins. Furthermore, we have mapped up to 908 unique sites with 300 μg of proteins (45 mg of placental tissue). With significantly more starting materials, the published study using human samples as starting material yielded 1,094 HexNAc sites using 600 μg of peptides (100 mg of brain tissue) (Wang et al. 2017), demonstrating that our protocol was highly efficient.

Previous landmark studies of the O-GlcNAcome have used murine synaptosome as starting materials. Using Wheat Germ Agglutinin combined with ETD-MS/MS, Burlingame and cohort identified 1,750 unique HexNAc sites using 30 mg of starting sample (Trinidad et al. 2012). Almost ten years later, using the PTM scan enrichment protocol, Myers and colleagues identified 1,000 unique sites with 4 mg of

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peptide digest from the same murine tissue (Burt et al. 2021). Both studies took advantage of fractionation before either enrichment protocol, thus decreasing the sample complexity and background, which differs from our approach. Our protocol includes depletion of abundant blood proteins and centrifugation followed by direct enrichment, which still yielded greater HexNAc site identification than the previous study, even with human banked tissues.

The reason why HexNAc is not directly assigned to GlcNAc in the O-GlcNAcome study, including ours, is that *O*-GlcNAc and *O*-GalNAc carry the same mass and are indistinguishable by MS. Extensive analysis of the Oxonium Ion for each PSM could differentiate between GlcNAc and GalNAc but is labor-intensive and time-consuming. In the interest of time, we took advantage of the differential cellular localization of the two glycosylation's. Indeed, *O*-GalNAcylation is the founding stone for complex *O*-glycosylation found in the extracellular compartment, either membrane-bound or secreted, and that utilizes the secretion pathway from ER to Golgi synthetic pathway. On the contrary, *O*-GlcNAcylation happens in the nucleus, cytosol, and mitochondria. Thus, we used this feature to sort the HexNAc peptides by cellular localization. We created a small bioinformatic tool, CytOVS, to fetch the cellular compartment information for each protein. We excluded the highly extracellular, ER- and Golgi-resident proteins from our pool of HexNAc peptides. We then classified the highly cytosolic, nuclear, or mitochondrial hexNAc proteins as O-GlcNAcylated. We then also provided a list of top *O*-GlcNAc targets that meet the following characteristics: (1) carry an HexNAc, (2) high cytosol, nuclear or mitochondrial localization, and (3) elevated PSM number. Each parameter can be adjusted per user need and provides a one-step process to extract the most probable *O*-GlcNAc target to follow up on. Using this stringent workflow, we confidently identified 407 unique *O*-GlcNac sites that could be prioritized for functional studies. Using a similar workflow, the study from Wang et al. using post-mortem brain tissue would have identified a little over 300 sites (Wang et al. 2017), and the original PTM scan study from Burt et al. would have confidently identified 68 *O*-GlcNAcylated proteins (Burt et al. 2021). However, we also acknowledge that the STRING database has more robust knowledge of human proteins than other organisms, such as mice, used in the previous PTM scan study.

The previous attempt to characterize the placental *O*-GlcNAcome has used mouse placenta as starting material (Qin et al. 2018). This study identified around 300 HexNAc sites on 76 proteins with 2 mg of starting mouse placental tissue using chemoenzymatic labeling. To date, no one has performed an *O*-GlcNAcomic analysis of the human placenta, which harbors key differences than the mouse placenta (Malassiné et al. 2003). For example, the human placenta secretes placental hormones unique to humans, such as placental GH, that are key to regulating women's metabolism during pregnancy. These hormones have been shown to play a critical role in gestational diseases only found in humans, such as gestational diabetes (Cox et al. 2009; Costa 2016), highlighting the importance of using human placentas to investigate *O*-GlcNAc in gestational diseases.

Interestingly, our study identifies essential insulin signaling effectors, such as IRS4 and PIK3C2G, as new *O*-GlcNAcylated proteins in the placenta that might be good targets to investigate in gestational diabetes. We also identified 82 *O*-GlcNAcylated proteins showing high placenta expression. These proteins were particularly prevalent in mast cell activation and linked to labor onset (Needham et al. 2016). This is likely correlated with the timing of retrieval for the starting material for this experiment, which is collected at delivery. The enrichment in cell death might also be a consequence of delivery. However, it has also been linked to placental diseases such as pre-eclampsia and might open new research avenues for this condition (Sharp et al. 2010). Cell adhesion has been highlighted in the previous mouse placenta study and our study (Qin et al. 2018). Overall, GO analysis of *O*-GlcNAcylated proteins in the human placenta highlights a key role for *O*-GlcNAc in the regulation of transcription, from DNA binding to repair and RNA stability, as previously highlighted in (Wulff-Fuentes et al. 2021). As a developmental organ, a special role for *O*-GlcNAcylation in the cell cycle was particularly emphasized in this study.

To conclude, mapping the human *O*-GlcNAcome in various tissues is an essential step toward creating targets for various human diseases. However, challenges in using human tissue as starting material include the limited quantity of tissues available and the contamination of the sample with blood contents. Stringent cleaning to deplete blood and extracellular proteins and adequate *O*-GlcNAc enrichment is critical to extracting meaningful data for an O-GlcNAcomic workflow from limited starting materials. Furthermore, a thorough sorting of the MS data can assist in identifying the most promising candidate to follow upon in vitro studies and improve our understanding of *O*-GlcNAc's role in health and diseases.

Material and methods

Frozen human placenta

Placenta samples were obtained from the Medical College of Wisconsin Maternal Research and Placenta and Cord Blood Bank (MCW MRPCB) under IRB approval. Written informed consent was received from patients before participation in this study. Healthy placentas without gestational complications were chosen for this pilot study ($n = 2$). Placentas are routinely flash-frozen within an hour of delivery. Samples for this study were recovered from frozen samples and shaved using a cryostat. Recovered placental shaves, yielded a total of 175 microns, equivalent to 45 mg of tissue.

Tissue lysis

Samples were lysed for 1 h at 4 °C in 200 μL of radioimmunoprecipitation assay lysis buffer (RIPA) (10 mM Tris-HCl, 150 mM NaCl, 1% Triton X-100 [v/v], 0.5% NaDOC [w/v], 0.1% sodium dodecyl sulfate [w/v], 10 μM PUGNAc, and EDTA-free Protease Inhibitor (Pierce), sonicated for 3 × 5 s pulses, and centrifuged 15 min at 18,000 × rcf at 4 °C; the supernatant was collected.

Chloroform/methanol protein precipitation

When needed, protein precipitation was performed by mixing 200 μL of samples with 600 μL of methanol, 150 μL of chloroform, and 400 μL of water and briefly vortexed. Samples were then centrifuged at 18,000 rcf for 5 min. The upper aqueous phase was removed, while the protein precipitate's interface layer was left intact. 450 μL of methanol was added and vortexed, followed by another centrifugation cycle. The supernatant was removed and discarded. Protein pellets were left to air-dry for 5 min, covered by a lint-free tissue.

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Blood protein cleanup

Samples were cleaned of major blood proteins using either the Human Serum Albumin (HSA)/ Immunoglobulins (Igs) or Top14 Abundant Protein Depletion Midi Spin Columns (Thermofisher Scientific). According to the manufacturer protocol, 100 μL of lysate was loaded, and proteins were recovered in about 800 μL of recovery buffer. Samples were precipitated or used directly as starting materials for the *O*-GlcNAc enrichment protocol (Cell signaling PTMscan).

SDS-PAGE and total protein stain

Laemmli Buffer was added to each protein lysate and boiled at 95 °C for 5 min before separation by SDS-PAGE. Samples were resolved on 10% or 4%–20% Tris-glycine gels. Proteins were transferred from gels onto nitrocellulose membranes using wet transfer electrophoresis at 4 °C for 14 h at 10 V. Membranes were washed 2 times for 2 min with ultrapure water, covered for 10 min in Invitrogen™ No-Stain™ Protein Labeling Reagent, and washed an additional 3 times with ultrapure water for 2 min. The membranes were then imaged using the Odyssey Fc imager (LI-COR) at 600 nm emission channel.

O-GlcNAc labeling

The Click-IT™ O-GlcNAc Enzymatic Labeling System (Thermofisher Scientific) was used to label *O*-GlcNAcylated protein according to the manufacturer's instructions. Labeled protein extracts were then clicked with the Click-IT™ Tetramethylrhodamine (TAMRA) Protein Analysis Detection Kit and loaded on SDS-page for visualization.

O-GlcNAcylated protein enrichment

The *O*-GlcNAc PTMscan kit (PTMScan *O*-GlcNAc [GlcNAc-S/T] Motif Kit #95220, Cell Signaling Technology, Inc) was used to enrich protein lysate for O-GlcNAc according to the original manufacturer protocol. For the optimized protocol, the following experimental details were followed: proteins were reduced and alkylated exactly as per protocol but were then not digested. Instead, we continued with denatured but undigested protein for the rest of the protocol. Instead of using C18 columns for in-between clean-up steps, we performed methanol-chloroform protein precipitation as described above. Samples were then dried using a speed vac and resuspended in a suitable buffer based on future application (SDS-PAGE or MS).

Mass spectrometry

Samples were suspended in 100 μL of 40% Invitrosol, 20% acetonitrile, and 100 mM ammonium bicarbonate. Two μg of trypsin was added, and the digest proceeded overnight at 37 °C. Following cleanup using the SP2 method, sample concentrations were measured using the Pierce Fluorometric Peptide Assay and were diluted to 100 ng/μL. Thermo Scientific Pierce Peptide Retention Time Calibration Mixture was added at 4 nM concentration, which diluted the samples to 25 ng/μL. Each sample was analyzed on a Thermo Scientific Orbitrap Fusion Lumos MS via two technical replicate injections in each method, one using HCD and product ion triggered EThcD, the other HCD and EThcD for all MS2 spectra. All MS data were analyzed using Proteome Discoverer 2.4 (Thermo) platform. See Supplementary Information for further details regarding the methods and data analysis. Graphs with annotated peptides were generated using the spectrum annotator web tool (Brademan et al. 2019).

Bioinformatic sorting of HexNAc proteins

Following MS, a dataset was generated, including proteins, their associated HexNAc counts, and peptide-spectrum match (PSM) data. We then used Cytoscape 3.10.1 (Shannon et al. 2003), integrating plugins from the STRING database (Szklarczyk et al. 2021), using the STRING app (Doncheva et al. 2023) and AutoAnnotate (Kucera et al. 2016). We automated the analysis workflow using the CyREST API (Ono et al. 2015) and the py4cytoscape 1.8.0 Python library. This automation was achieved through a custom script named CytOVS. We have released CytOVS with a straightforward graphical user interface for convenience. In brief, proteins in the dataset are cross-referenced with the STRING database. The script is designed to classify proteins as intracellular or extracellular based on a predefined set of compartment location data and user-set thresholds. This process retrieves details, including each protein's tissue and cellular compartment locations, quantified as enrichment factors.

Additionally, we integrated HexNAc and PSM data to categorize proteins into distinct groups: those without HexNAc modification, those with unidentified HexNAc, extracellular HexNAc, *O*-GlcNAcylated proteins, and top *O*-GlcNAc targets. After processing, the data are re-imported into Cytoscape and presented in a group-attribute layout. This layout explicitly highlights the dataset's distribution, focusing on identifying *O*-GlcNAcylated proteins and the most significant targets. An example of a network generated with this app is available as a supplementary document. We have made the source code for CytOVS and its GUI version available to the community (see github.com/synthaze/cytovs). A Tutorial Video that explains how to install and run CytOVS is available as Supplementary material.

Gene ontology analysis

GO was performed using Cytoscape and the ClueGO app. The functional analysis mode was used with "GO: biological processes" as ontology terms.

Software, databases, and data repository

Statistics and graphics were alternatively done on Prism GraphPad and R. Databases used in this study include UniprotKB (https://www.uniprot.org/), The *O*-GlcNAc Database (https://www.oglcnac.mcw.edu) and GlyGen (https://www.glygen.org/). Proteomic data are available on the PRIDE public repository under the reference PXD047766. Figures were created with Biorender. We have made the source code for cytovs and its GUI version available to the community (see github.com/synthaze/cytovs).

Acknowledgments

We would like to thank the MS Core for their help with generating the MS data. We thank the OVS lab for their comments and feedback on this project. Finally, we thank Dawn Wenzel, PhD, for her feedback on the manuscript.

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Author contributions

Taxonomy Sarai Luna (Conceptualization [Equal], Data curation [Equal], Formal analysis [Equal], Methodology [Equal], Validation [Equal], Writing—original draft [Equal], Writing—review & editing [Equal]), Florian Malard (Conceptualization [Equal], Data curation [Equal], Software [Equal], Visualization [Equal], Writing—review & editing [Equal]), Michaela Pereckas (Formal analysis [Equal], Methodology [Equal], Validation [Equal]), Mayumi Aoki (Data curation [Equal], Formal analysis [Equal], Methodology [Equal], Validation [Equal]), Kazuhiro Aoki (Data curation [Equal], Formal analysis [Equal], Methodology [Equal], Validation [Equal]), Stephanie Olivier-Van Stichelen, (Conceptualization [Equal], Funding acquisition [Lead], Investigation [Lead], Project administration [Lead], Resources [Equal], Software [Equal], Supervision [Lead], Validation [Equal], Visualization [Equal], Writing—original draft [Equal], Writing—review & editing [Equal])

Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

Funding

Eunice Kennedy Shriver National Institute of Child Health and Human Development, (Grant / Award Number: 'R01HD104808'). Cardiovascular center Medical College of Wisconsin, ('Sally Bentley Early Stage Investigator Pilot Award')

Conflict of interest statement: None declared.

Data availability

Proteomic data are available on the PRIDE public repository under the reference PXD047766. Figures were created with Biorender. We have made the source code for cytovs and its GUI version available to the community (see github.com/synthaze/cytovs).

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